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Techno-economic evaluation of a thermochemical waste-heat recuperation system for industrial furnace application: Operating cost analysis

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ABSTRACT

An operational cost analysis was conducted to determine the optimal parameters for achieving maximum net present profit value through thermochemical recuperation (TCR). The parameters of the TCR system (enthalpy, temperatures of exhaust gas and synthesis gas) were determined based on computational fluid dynamics (CFD) modeling. The results of the CFD modeling were verified using experimental data. The techno-economic evaluation of the TCR system was performed for two types of industrial furnaces with methane mass flow rate of 1.0 kg/s: a glass-melting furnace with an exhaust gas temperature of 1500 °C and a forging furnace with an exhaust gas temperature of 900 °C. The operational parameters of the steam methane reforming process varied during analysis: a residence time of 50–250 $kg_{cat}\cdot s/mol_{CH_4}$, a feed steam-to-methane ratio of 1.0–3.0 mol/mol, and an operational pressure of 10 bar. Various prices for catalyst, electricity, and natural gas, as well as different replacement periods for the catalyst, were analyzed. It was found that the net present profit had an extreme value that depended on operational and cost parameters. The optimal residence time (specific mass of catalyst per mole flow rate of methane) was determined for various operational and cost parameters of the TCR system.

1. Introduction

Transitioning to carbon-free energy and industry is intrinsically linked with improving energy efficiency. To decarbonize the industry, we need to decarbonize its energy sources [1]. There are areas of the industry where abandoning hydrocarbon fuels in the coming years is not feasible, such as metallurgical and glass furnaces [2,3]. One way to reduce carbon dioxide emissions in these cases is to improve the energy efficiency of such furnaces. Despite the significant efforts made by scientists and engineers in recent years, there is still a high potential for increasing the energy efficiency of high-temperature furnaces because the exhaust gas temperature can exceed 1000 °C [4,5]. The current state-of-the-art for such furnaces focuses on using air preheaters to recuperate exhaust heat and to heat air before combustion [6,7]. There are various approaches for using this kind of exhaust heat recuperation, each with its own advantages and disadvantages.

In the last few decades, different research groups have been studying thermochemical waste-heat recuperation as a promising way to improve the energy efficiency of high-temperature furnaces [8–11]. The main concept of thermochemical recuperation involves using exhaust heat for endothermic fuel transformation to obtain hydrogen-rich fuel, which is then used as a fuel for industrial furnaces. The details of this concept are provided in the authors' previous publications [12,13].

One of the first industrial applications of thermochemical waste heat recuperation for glass melting furnaces was implemented by Pereletov's research group from the Moscow Power Engineering Institute in 1978 [14]. The scientists and engineers developed and implemented a thermochemical waste heat recuperation system based on steam methane reforming to improve the energy efficiency of glass melting furnaces. The tests showed that the specific fuel consumption per 1 kg of charge was reduced from 0.74 to 0.524 m^3, i.e. by 29.2%, or 1.41 times, for the glass melting furnace without and with thermochemical waste heat recuperation, respectively. The authors also reported that the specific fuel consumption per 1 kg of fiberglass was reduced from 1.34 to 0.994 m^3, i.e. by 25.8%, or 1.35 times, for the furnace without and with thermochemical waste heat recuperation, respectively.

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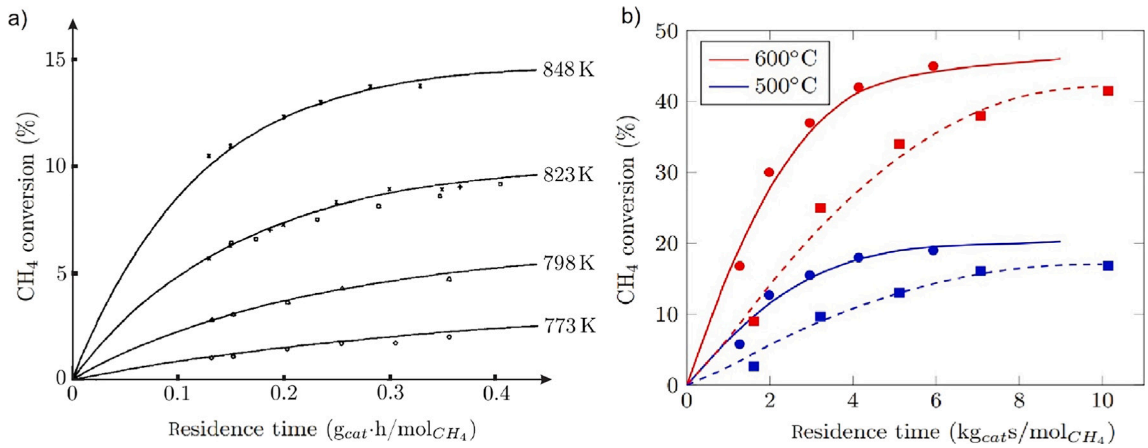

Fig. 1. Methane conversion vs. residence time in steam methane reforming: (a) Intrinsic kinetic study conducted by Xu and Froment [20]; (b) Catalyst packed bed study conducted by Pashchenko [21].

In the 1980s, the Dutch VEG-Gas Institute and TNO (Technical Netherlands Organisation) reported the results of industrial tests of the thermochemical waste heat recuperation (TCR) system based on steam methane reforming for glass-melting furnaces [15,16]. The TCR system was used to recuperate waste heat after furnaces with temperatures of about 1400 °C. The TCR system consists of a reformer, air heater, reaction mixture heater, and boiler. Synthesis gas after the reformer was fed for combustion to the furnace as a fuel. The authors showed that up to 70% of waste heat was recuperated while the air temperature was about 700 °C. Moreover, they showed that NO_x emissions were decreased compared to the traditional recuperation system based on only air heating.

Researchers from GTI – Gas Technology Institute published their report on the application of thermochemical waste heat recuperation to improve the energy efficiency of glass furnaces [17]. The authors compared the results of numerical simulation with experimental results from the pilot-scale thermochemical recuperation system (furnace energy input 150 kW). The tests made it clear that the use of thermochemical waste heat recuperation leads to an increase in energy efficiency of up to 30% compared to the traditional recuperation system based on air heating.

Recently, researchers from the Graz University of Technology published a series of articles on thermochemical waste-heat recuperation [10,18,19]. Gaber et al. presented the results of experimental tests of a TCR system based on natural gas reforming using various oxidizers, such as steam-flue gas, steam-flue gas-oxygen, and steam-oxygen mixtures [10]. The authors mainly focused on the parameters of the methane reforming process and studied the effect of operational parameters on reforming efficiency. Additionally, they conducted experimental tests on carbon formation conditions, which is important for TCR systems as carbon deposition leads to catalyst poisoning.

The authors of this paper were actively involved in research on TCR. Pashchenko provided the first law energy analysis of a thermochemical waste heat recuperation system to understand the inputs and outputs of the energy balance [8]. It was confirmed that the maximal energy efficiency of the TCR system can be achieved when high-temperature exhaust gas is utilized because, in this case, the maximal methane conversion can be achieved. Pashchenko and Karpilov reported that the maximal efficiency of gas turbines is observed for high-temperature exhaust heat [22]. They also showed that the efficiency of the TCR system decreases when the operating pressure increases because higher pressure gives lower methane conversion and, as a result, a lower amount of exhaust heat is converted into the chemical energy of new synthetic gas. The comprehensive overview of the use of the thermochemical waste-heat recuperation systems are provided in Pashchenko's review [23].

Popov et al. provided a thermodynamic analysis of an industrial glass-melting furnace to evaluate various configurations of thermochemical waste-heat recuperation systems [9]. They analyzed various oxidizers for natural gas in the reformer, including steam, flue gas, and steam/flue gas mixture, under various operational parameters. The authors showed that the fuel consumption and costs of primary energy are non-monotonic and have an extreme that depends on operational parameters. Based on calculations, the authors confirmed that using a TCR system leads to an increase in energy of up to 10%–25% without affecting other parameters of the process.

In the above-mentioned investigations, thermochemical waste-heat recuperation is based on steam methane reforming (SMR). The efficiency of SMR depends on various variables such as operational (temperature, pressure, inlet gas composition, residence time) and design (type of catalyst, reformer, shape of catalyst). Therefore, the efficiency of thermochemical recuperation is also affected by those operational and design variables. There are no publications on the techno-economic evaluations of the thermochemical waste-heat recuperation systems for industrial furnace application. In this paper, we determined the effect of operational and design parameters on the economic efficiency of TCR. Moreover, in this paper, we took into account an energy demand to compensate for the pressure drop in the reformer. Therefore, techno-economic evaluation was performed for a wide range of practically interesting cases.

2. Methodology

In the thermochemical waste-heat recuperation system, waste heat is converted into chemical energy of hydrogen-rich fuel due to primary fuel (e.g. methane) thermochemical transformation. As a result, the amount of waste heat converted into chemical energy is in direct ratio to fuel conversion degree (e.g. methane conversion). In other words, greater methane conversion results in a greater amount of waste heat being converted into chemical energy. The detailed clarification of main principles of thermochemical waste-heat recuperation is provided in the authors' previous publications [8,24].

A thermochemical reformer is a core element of the thermochemical waste-heat recuperation system. The chemical reactions take place in a packed bed filled with catalysts. For industrial-scale reformers, Ni-based catalysts are widely used [25–27]. The steam methane reforming process is a well-studied process, and there is sufficient information about the effect of operational and design variables on methane reforming performance. In general, the industrial-scale chemical reformer is a heat exchanger with a reaction space filled with catalyst particles. The price of Ni-based catalysts varies from 20 to 200 USD per 1 kg [28].

According to the authors' experiments as well as experiments and numerical modeling performed by other authors, methane conversion

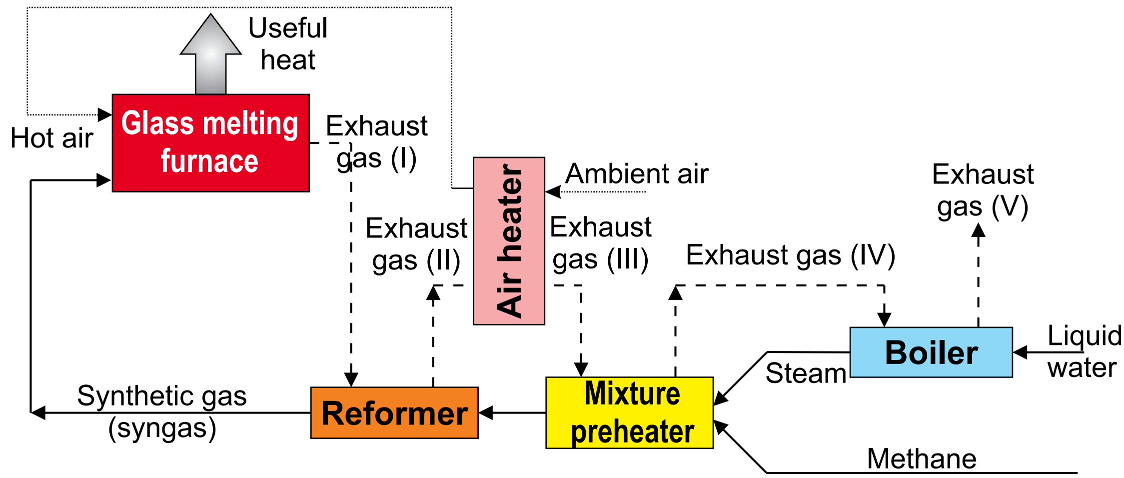

Fig. 2. Schematic diagram of a glass-melting furnace with thermochemical waste-heat recuperation.

is non-linearly dependent on the catalyst mass (residence time - mass of catalyst per mole of methane per second) in the reformer. The dependence of methane conversion on catalyst mass is close to logarithmic. Schematically, the dependence of methane conversion on residence time (specific mass of catalyst per methane mole flow rate) is shown in Fig. 1a. Evidently, the catalyst near the reformer inlet results in greater methane conversion than the catalyst near the outlet.

Based on data from Fig. 1, it can be concluded that economic efficiency of the use of thermochemical waste heat recuperation depends on a mass of catalyst (residence time) filled in the reformer. Moreover, an increase in packed bed length (mass of catalyst) leads to an increase in pressure drop. To compensate the additional pressure drop, electricity should be additionally consumed.

In this paper, we analyzed the operating expenses (OPEX) as a function of mass of catalyst (residence time) for given initial data related to real glass melting furnace and forging furnace. The schematic diagram of a glass melting furnace with thermochemical waste heat recuperation is presented in Fig. 2.

In the schematic diagram from Fig. 2, exhaust gas sequentially passes a reformer, air heater, steam-methane mixture preheater, and boiler (steam generator). The core element of recuperation system is steam methane reformer filled with the Ni-based catalysts. In the reformer, synthesis gas (syngas) are produced and further this syngas is used as a fuel. In the reformer, the steam methane reforming process are taking place. The overall process may include the different chemical reactions. However, there is sufficient information that only three reactions have significant effect on material and energy balance [20,29]:

$$CH_4 + H_2O \Leftrightarrow CO + 3H_2 + 206.1\ \text{kJ/mol}; \tag{1}$$

$$CO + H_2O \Leftrightarrow CO_2 + H_2 - 41.1\ \text{kJ/mol}; \tag{2}$$

$$CH_4 + 2H_2O \Leftrightarrow CO_2 + 4H_2 + 165.0\ \text{kJ/mol}. \tag{3}$$

2.1. Cost evaluation model

Cost evaluation calculations of the TCR system are performed based on initial data presented in Table 1. In this article, the cost evaluations problem is reduced to determining the optimality criterion. In this study, the economic effect (net present profit) of using the TCR system was chosen as an optimality criterion. In general, the economic effect (net present profit) of using the TCR system is the difference between the cost of recuperated (saved) energy and operating expenses (OPEX) [30]:

$$C_{tot} = B - OPEX_{tot} = \frac{H_{rec}}{LHV_{ng}} \cdot C_{ng} - \sum OPEX_i \tag{4}$$

where B – cost of recuperated (saved) energy due to using TCR system, \$; $OPEX_{tot}$ – total operating expenses for TCR system, \$; H_{rec} – recuperated heat in TCR system, MW; LHV_{ng} – lower heating value of natural gas, MJ/kg; C_{ng} – price of natural gas, \$/kg; $OPEX_i$ – ith operating expense, \$.

In this paper, we do not consider the capital expenditures for metal of heat exchanger (reformer) and other terms because for hour- or day-scale the effect of capital expenditures to overall economic balance is negligible.

Recuperated heat can be determined from heat balance as the following:

$$H_{rec} = H_{exh} - H'_{exh} = \Delta H_{smr} + \Delta H_{sg} \tag{5}$$

where H_{exh}, H'_{exh} – enthalpy of exhaust gas before and after TCR system, MW; ΔH_{smr} – enthalpy of the steam methane reforming process, MW; ΔH_{sg} – change of enthalpy of synthesis gas in TCR system, MW.

In this study, the terms for recuperated heat from Eq. (5) was determined based on CFD-modeling described below. Based on author's previous studies, it can be concluded that recuperated heat non-linear depends on mass of catalyst (residence time) in TCR system (Fig. 1). Moreover, an increase in a mass of catalyst leads to an increase in reformer length, and, as a result, in pressure drop. In summary, the operating expenses can be determined as the following:

$$OPEX_{tot} = OPEX_{cat} + OPEX_{el} \tag{6}$$

where $OPEX_{cat}$ – operating expense for catalyst, \$; $OPEX_{el}$ – operating expense for electricity demand, \$.

The Ni-based catalysts are widely used for steam methane reforming. There is sufficient information from industry and literature that catalytic activity is decreasing on time. Therefore, the catalysts should be replaced to obtain high constant activity. In industry, the replacement period varies from a few weeks up to a few years. This period depends on the type of catalyst and operating regimes. Therefore, in this paper, we analyzed various periods of catalyst replacement.

The initial data for the cost evaluation model are summarized in Table 1. Table 1 contains two type of initials data: constant and variable which are changed in the model.

In this paper, exhaust gas temperature of 900 °C and 1500 °C is considered for operating cost analysis because those temperature are typical exhaust gas temperatures for a forging furnace and glass-melting furnace, respectively.

To obtain the dependence of methane conversion and recuperated heat in the reformer on mass of catalyst, CFD-modeling was conducted.

Table 1
Initial data for the cost evaluation model.

Parameter	Unit	Value
Fuel consumption	kg/s	1.0
Exhaust gas temperature	°C	900; 1500
Reformer tube length	mm	3000
Reformer tube diameter	mm	100
Wall thickness	mm	3
Catalyst type		Ni-Al_2O_3
Catalyst particle size (sphere)	mm	20
H_2O:CH_4 molar ratio	–	1.0; 2.0; 3.0
Catalyst replacement period	month	3–24

Table 2
Governing equations for CFD-model.

Equation	Formula
Mass conservation	$\nabla(\rho\vec{v}) = 0$
Momentum conservation	$\nabla(\rho\vec{v}\vec{v}) = -\nabla p + \nabla(\bar{\bar{\tau}}) + \rho\vec{g} + \vec{F} - S_i$
Energy conservation	$\nabla\vec{u}(p + \rho h_f) = \nabla\left[k_{eff}\nabla T - \left(\sum h_j\vec{J}_j\right) + \bar{\bar{\tau}}\vec{v}\right] + S_f^h$
Mass transfer	$\nabla(\rho\vec{v}Y_j) = -\nabla\vec{J}_j + R_j$
Chemical kinetics	Hou&Hughes' kinetics [36]

2.2. CFD-modeling

In general, the methane reformer consists of many elementary tubes filled with catalysts [31]. In this study, CFD modeling of a single reformer tube was conducted to understand the effect of catalyst mass (residence time) on methane conversion and pressure drop. Ansys Fluent software was used for numerical calculation. The residence time changes by varying the mass flow rate of the reaction mixture in the reformer tube. The number of reformer tubes is determined as a function of the mass flow rate of the reaction mixture. For higher velocity, the number of reformer tubes is lower; for lower velocity, the number of reformer tubes is higher. It is expected that methane conversion and pressure drop will be the same for all elementary reformer tubes. For the cost evaluation model, the results obtained via CFD modeling were scaled up to the real size of the reformer.

A pseudo-homogeneous model is used for CFD modeling of steam methane reforming in a packed bed. In this model, the packed bed is presented as a porous medium, and reacting flow through a packed bed is simulated as flow through a porous medium. This approach for CFD modeling of SMR was implemented in previous papers published by the authors of this paper [32] as well as in articles presented by other authors [33–35].

The governing equations for CFD-model include the mass, momentum, and energy conservation equations accompanied by chemical kinetic model (see Table 2).

The diffusion flux of jth component can be expressed as:

$$\vec{J}_j = -\left(\rho \cdot D_{j,m} + \frac{\mu_t}{Sc_t}\nabla\right) Y_j - D_{T,j}\frac{\nabla T}{T} \tag{7}$$

where $D_{j,m}$ and $D_{T,j}$ – mass diffusion coefficient and thermal diffusion coefficient for j species, respectively; Sc_t and μ_t – turbulent Schmidt number and turbulent viscosity, respectively.

The packed bed is modeled as a porous medium using term S_i in the momentum equation. S_i is defined as the following:

$$S_i = -\left(\frac{\mu}{\alpha}v_i + C_2\frac{1}{2}\rho|v|v_i\right) \tag{8}$$

The coefficients from Eq. (8) α (permeability) and C_2 (inertial resistance factor) were defined from the Ergun equation [37,38]:

$$\alpha = \frac{D_p^2}{150}\frac{\epsilon_{ip}^3}{(1-\epsilon_{ip})^2};\ C_2 = \frac{3.5}{D_p}\frac{(1-\epsilon_{ip})}{\epsilon_{ip}^3}. \tag{9}$$

where D_p – catalyst particle diameter; ϵ_{ip} – packed bed porosity.

Table 3
The coefficients for the reaction rates of kinetic model.

Reaction rate		
k_1	=	$5.922 \cdot 10^8 \exp\left(\frac{-209\,200}{R_g T}\right)$
k_2	=	$6.028 \cdot 10^{-4} \exp\left(\frac{-15\,400}{R_g T}\right)$
k_3	=	$1.093 \cdot 10^3 \exp\left(\frac{-109\,400}{R_g T}\right)$
Adsorption		
K_{H_2O}	=	$9.251 \cdot \exp\left(\frac{-15\,900}{R_g T}\right)$
K_{H_2}	=	$5.68 \cdot 10^{-10} \exp\left(\frac{93\,400}{R_g T}\right)$
K_{CO}	=	$5.127 \times 10^{-13} \exp\left(\frac{140\,000}{R_g T}\right)$
Equilibrium		
K_1	=	$1.198 \cdot 10^{17} \exp\left(\frac{-26\,830}{T}\right)$
K_2	=	$1.767 \cdot 10^{-2} \exp\left(\frac{4400}{T}\right)$
K_3	=	$2.117 \cdot 10^{15} \exp\left(\frac{-22\,430}{T}\right)$

The effective thermal conductivity k_{eff} in the energy conservation equation is calculated from the following expression for porous medium:

$$k_{eff} = \epsilon_{ip}k_f + (1-\epsilon_{ip})k_s, \tag{10}$$

where k_f, k_s – fluid phase and solid medium thermal conductivity, respectively.

To solve RANS equation, RNG k-ϵ model is used [39].

The SIMPLE algorithm is used to establish a relationship between velocity and pressure corrections to enforce mass conservation and to obtain the pressure field.

The first-order upwind discretization is chosen [37].

Standard interpolation scheme is used because for most cases the "standard" scheme is acceptable. In our simulation the pressure drop is insignificant and computational domain is simple.

The P-1 radiation model is used for radiation simulation [40].

The reaction rates of the chemical reactions (1)–(3) are determined based on the kinetic model presented by Hou and Hughes [36]:

$$R_1 = \frac{k_1(p_{CH_4}p_{H_2O}^{0.5} - p_{H_2}^3 p_{CO}/K_1 p_{H_2O}^{0.5})}{p_{H_2}^{1.25} \cdot \mathrm{DEN}^2} \tag{11}$$

$$R_2 = \frac{k_2(p_{CO}p_{H_2O}^{0.5} - p_{H_2}p_{CO_2}/K_2 p_{H_2O}^{0.5})}{p_{H2}^{0.5} \cdot \mathrm{DEN}^2} \tag{12}$$

$$R_3 = \frac{k_3(p_{CH_4}p_{H_2O} - p_{H_2}{}^4 p_{CO_2}/K_3 p_{H_2O})}{p_{H_2}^{3.5} \cdot \mathrm{DEN}^2} \tag{13}$$

$$\mathrm{DEN} = 1 + K_{CO} \cdot p_{CO} + K_H \cdot p_{H_2}^{0.5} + K_{H_2O} \cdot p_{H_2O}/p_{H_2} \tag{14}$$

The coefficients for (11)–(13) are presented in Table 3:

The kinetic model developed by Hou and Hughes was developed for intrinsic kinetics for the case when no diffusion limitation inside catalyst particle. According to Table 1, the shape of catalyst particle is sphere and diameter is 20 mm. Therefore, intra-particle diffusion limitation is taking place. Intra-particle diffusion limitation in steam methane reforming over Ni-based catalyst was recently assessed by Pashchenko [41]. Intra-particle diffusion limitation is expressed via an effectiveness factor:

$$\eta = \frac{R_{real,j}}{R_{intr,j}} \tag{15}$$

where $R_{real,j}$ – a reaction rate with diffusion limitation; $R_{intr,j}$ – a reaction rate without diffusion limitation (intrinsic kinetic).

The effectiveness factor is dependent on a particle size, temperature and steam-to-methane ratio according to the following equation [41]:

$$\eta = f_d(d) \cdot f_T\left(\frac{T}{1000}\right) \cdot f_b\left(\frac{N_{H_2O}}{N_{CH_4}}\right), \tag{16}$$

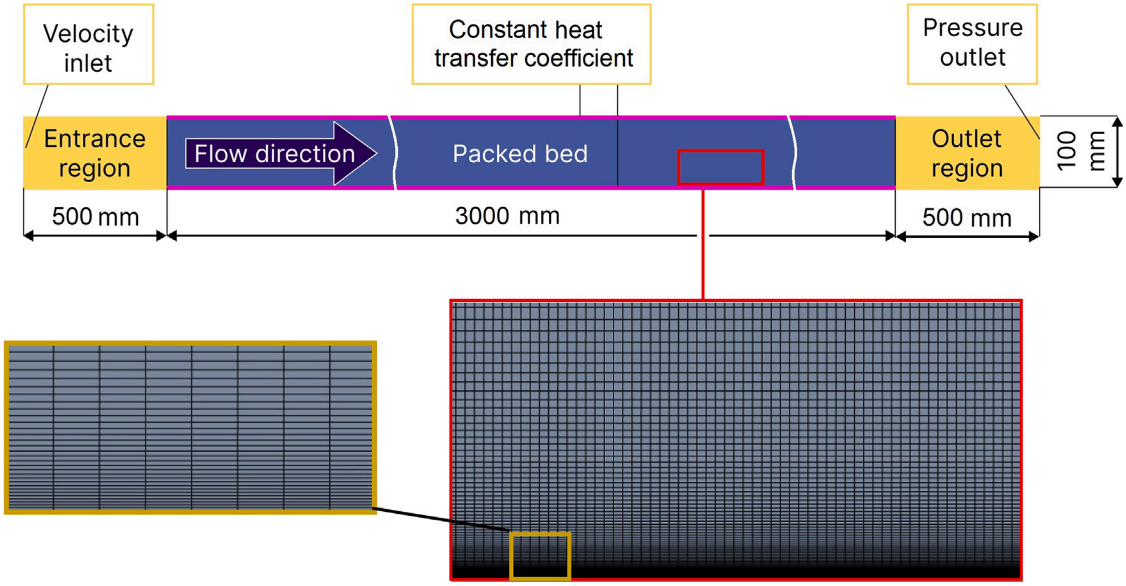

Fig. 3. Computational domain and mesh structure of the reaction space of reformer tube.

where $f_d(d)$ – the function of the effectiveness factor on diameter; $f_T\left(\frac{T}{1000}\right)$ – the function of the effectiveness factor on temperature; $f_b\left(\frac{N_{H_2O}}{N_{CH_4}}\right)$ – the function of the effectiveness factor on steam-to-methane ratio.

The functions $f_d(d)$, f_T, f_b from Eq. (16) were obtained from Pashchenko's paper [41]:

$$f_d(d) = 0.11/d; \tag{17}$$

$$f_T\left(\frac{T}{1000}\right) = -1.38875\cdot\left(\frac{T}{1000}\right)^2 + 1.90065\cdot\left(\frac{T}{1000}\right) + 0.368183; \tag{18}$$

$$f_b\left(\frac{N_{H_2O}}{N_{CH_4}}\right) = 1.10535 - 0.0966\cdot\left(\frac{N_{H_2O}}{N_{CH_4}}\right) \tag{19}$$

To adopt kinetic model for CFD simulation, the above mentioned equations was written using C++, and then the code was compiled and implemented to CFD-model via user-defined function (UDF) $DEFINE\_VR\_RATE$.

As a result of CFD-modeling, the following parameters and dependence as a function of residence time were determined:

- temperature distribution along reformer of exhaust and synthesis gas;
- synthesis gas composition;
- recuperated heat in a reformer;
- pressure drop.

The computational domain of the inner space and mesh structure is presented in Fig. 3. The length of the reformer tube is 4000 mm, with inlet and outlet zones (no catalyst) of 500 mm each and a reaction space of 3000 mm. The walls of the inlet and outlet zones are adiabatic. These zones are added to obtain a stable solution and to get a developed flow at the packed bed inlet. The diameter of the reformer tube is 100 mm. The number of mesh elements was determined based on a mesh independence study, and the total number of elements is 2,457,019. Synthesis gas temperature was chosen as a control parameter for the mesh independence study.

The main assumptions are provided below:

- The change in residence time is achieved via a change in methane mole flow rate;
- The heat transfer coefficient from the flue gas side is constant along the reformer wall;
- The heat transfer coefficient from the flue gas side is separately calculated and then integrated into the present CFD model;
- The temperature of reaction mixture at the reformer inlet is 400 °C;
- The operation pressure is 10 bar;
- The mass flow rate of flue gas corresponds to the mass flow rate of combustion products obtained by combustion of synthesis gas at an air equivalence ratio of 1.0.

A mathematical formulation of the boundary conditions can be expressed as the following:

– reformer inlet: $T_{in} = T_{in}^{gas}$; $Y_{i,in} = Y_{i,in}^{gas}$;
– reformer outlet: $\frac{\partial Y_i}{\partial x} = 0$; $\frac{\partial T^{gas}}{\partial x} = 0$.

3. Results and discussion

3.1. Heat and mass transfer

To verify our CFD model, the experimental results from Hou and Hughes' paper was used [36]. The CFD model was adopted for their experiments, where the reformer was filled with catalyst particles of 0.15–0.20 mm, considered as spheres. Additionally, the catalyst particles were diluted with inert particles of the same diameters. To account for this, we involved the correlation factor (γ) for the kinetic equations from Table 3 in our model:

$$\gamma = \frac{M_{cat}}{M_{total}} \tag{20}$$

where M_{cat} is a mass of the catalyst particles; M_{total} is a total mass of the active and inert particles.

Fig. 4a compares the numerical results and experimental data from Hou and Hughes' paper [36] for operational parameters of p = 300 kPa and CH_4:H_2O:H_2 = 1:5:1, with residence times varied by molar flow rates. Fig. 4a shows good correlation between the numerical and experimental results. The maximum deviation is 4.91% while the standard deviation is 3.86%.

Fig. 4b shows contours of CH_4 molar fraction, temperature, and velocity at 796 K, p = 300 kPa, CH_4:H_2O:H_2 = 1:5:1, and a residence time of 12 000 kg_{cat} s/$kmol_{CH_4}$. The temperature distribution does not have a notable effect on CH_4 conversion because the temperature deviation inside the reaction zone is less than 1 K, which is in agreement

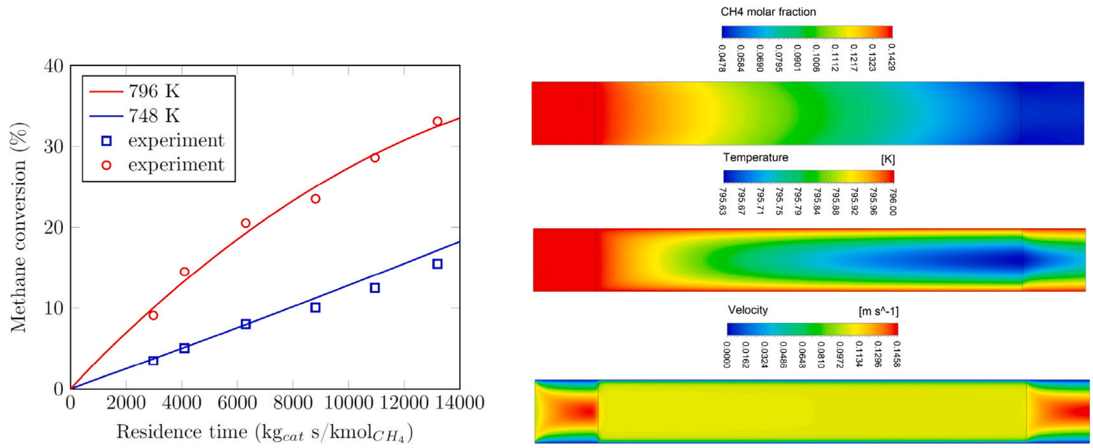

Fig. 4. (a) CH_4 conversion vs. residence time: solid lines – CFD modeling, hollow points – experimental results [36]; (b) Contours of CH_4 mole fraction, temperature and velocity at 796 K, p = 300 kPa, CH_4:H_2O:H_2 = 1:5:1, residence time 12 000 kg_{cat} s/$kmol_{CH_4}$.

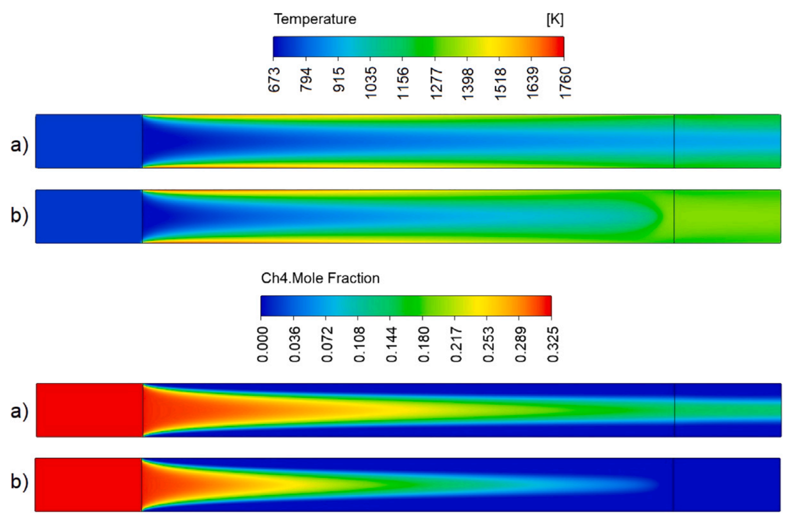

Fig. 5. Contours of temperature and CH_4 mole fraction in the reaction space of reformer tube for various residence time: (a) 150 $kg_{cat}s/mol_{CH_4}$; (b) 250 $kg_{cat}s/mol_{CH_4}$.

with Hou's paper [36] where the authors stated that the flow inside the reaction zone can be considered as isothermal flow. Fig. 4 confirms that the CFD model based on Hou and Hughes' kinetics can be used for numerical simulation of steam methane reforming to obtain accurate results.

The CFD model was applied to determine the parameters of synthesis gas at the reformer tube outlet as well as the temperature and molar fraction distributions along the reaction space for the reformer presented in Fig. 3. Fig. 5 depicts the temperature and CH_4 molar fraction distribution (scaled in radial and axial directions) in the reaction space of the reformer tube for H_2O:CH_4 = 2:1.

Fig. 5 shows that a decrease in residence time leads to a decrease in methane conversion, which is consistent with experimental data and numerical results of other researchers [36,42–45]. For instance, the methane conversion for 250 $kg_{cat}s/mol_{CH_4}$ is 0.97, while for 150 $kg_{cat}s/mol_{CH_4}$ and 50 $kg_{cat}s/mol_{CH_4}$, it is 0.85 and 0.51, respectively. However, an increase in residence time leads to only a slight change in outlet temperature. For example, at 250 $kg_{cat}s/mol_{CH_4}$, the outlet temperature is 974.8 °C, while at 150 $kg_{cat}s/mol_{CH_4}$ and 50 $kg_{cat}s/mol_{CH_4}$, it is 932.8 °C and 847.9 °C, respectively. This can be explained by the fact that a decrease in residence time leads to a decrease in CH_4 conversion and, as a result, to a decrease in the enthalpy of the steam methane reforming reaction. Therefore, less heat is consumed to accomplish steam methane reforming.

The CFD model was used to determine the parameters of the thermochemical recuperation system. Fig. 6 shows the average temperature at the reformer outlet for synthesis gas and exhaust gas. The variables on these plots are residence time and steam-to-methane ratio.

Fig. 6 illustrates that the character of the dependences for T_{exh} = 900 °C and T_{exh} = 1500 °C is different. The main reason for this observation is the effect of the process temperature on methane conversion. A decrease in temperature leads to a decrease in methane conversion. At the same time, a decrease in methane conversion leads to a decrease in the enthalpy of steam methane reforming reaction. An increase in steam-to-methane ratio leads to a decrease in syngas temperature. There are two main reasons for this observation. First, an increase in steam-to-methane ratio leads to an increase in CH_4 conversion and an increase in the reaction-mixture mass flow rate through a reformer tube. Second, an increase in steam-to-methane ratio leads to a decrease in exhaust gas temperature at a reformer outlet because an increased reaction mixture mass flow rate leads to an increase in heat transfer coefficient.

Fig. 7 depicts the effect of steam-to-methane ratio and residence time on CH_4 conversion. For T_{exh} = 900 °C, the dependence of CH_4 conversion on residence time is almost linear, while for T_{exh} = 1500 °C, it is close to logarithmic. This observation is due to the fact that for T_{exh} = 1500 °C, CH_4 conversion reaches its equilibrium value at a residence time of about 200–250 $kg_{cat}s/mol_{CH_4}$ for the given

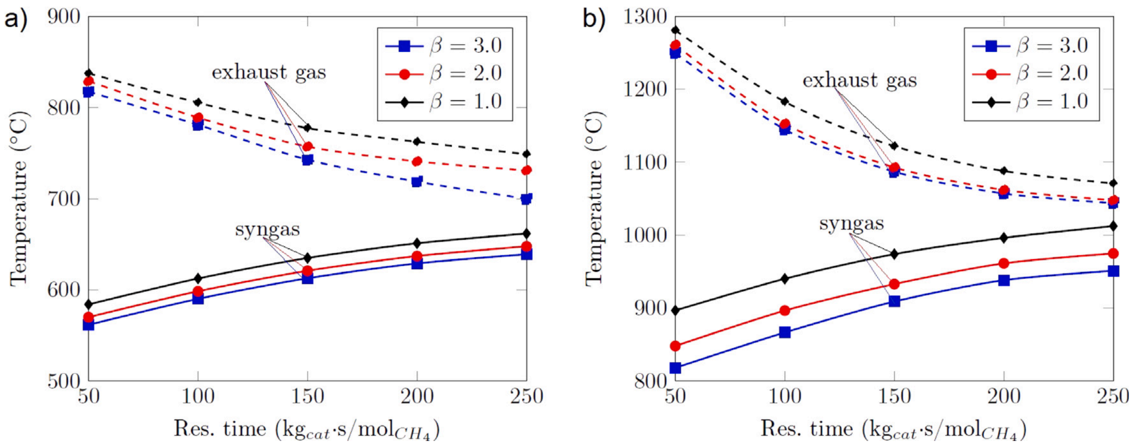

Fig. 6. Average temperature at reformer outlet as a function of residence time for various exhaust gas temperatures at reformer inlet: (a) T_{exh} = 900 °C; (b) T_{exh} = 1500 °C.

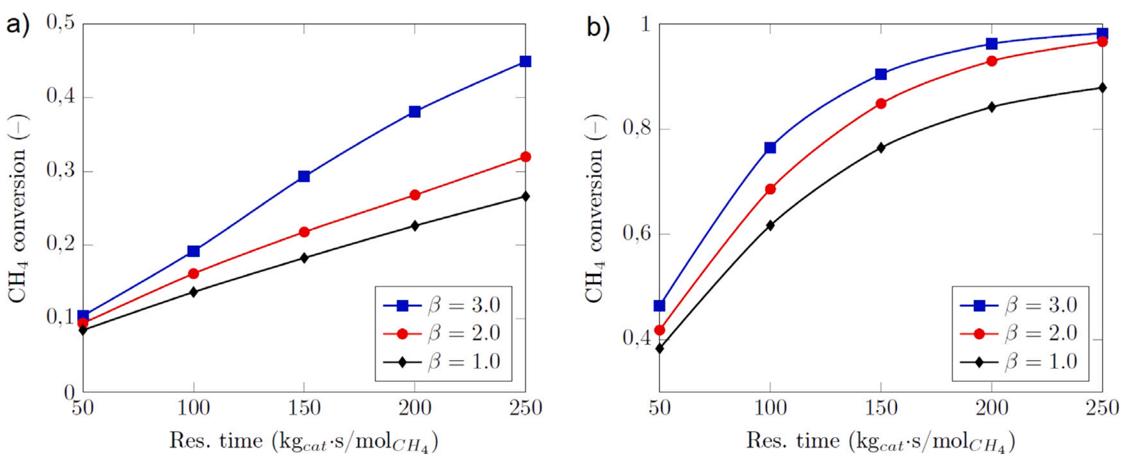

Fig. 7. CH_4 conversion as a function of residence time for various exhaust gas temperatures at reformer inlet: (a) T_{exh} = 900 °C; (b) T_{exh} = 1500 °C.

temperature levels, while for T_{exh} = 900 °C, CH_4 conversion is far from its equilibrium value.

The key parameter of the efficiency of the thermochemical recuperation system is recuperated heat. Recuperated heat in the thermochemical recuperator-reformer can be calculated as a sum:

$$H_{rec} = \Delta H_{smr} + \Delta H_{sg} \tag{21}$$

Fig. 8 shows the effect of residence time and steam-to-methane ratio on recuperated heat for a methane mass flow rate of 1 kg/s. An increase in residence time leads to a linear increase in recuperated heat for T_{exh} = 900 °C and β = 1.0 and β = 2.0. However, for T_{exh} = 900 °C and β = 3.0 and for all steam-to-methane ratios at T_{exh} = 1500 °C, recuperated heat is nonlinearly dependent on residence time. For example, for T_{exh} = 1500 °C and β = 3.0, the first 100 $kg_{cat}s/mol_{CH_4}$ of residence time gives about 16 MW of recuperated heat, while the last 100 $kg_{cat}s/mol_{CH_4}$ gives about 2 MW of recuperated heat. Therefore, a linear increase in residence time gives a nonlinear increase in recuperated heat, and techno-economic evaluation of residence time for various operational parameters is a crucial task.

3.2. Operational costs

The aforementioned results show that temperature and synthesis gas composition depend on residence time and other operational parameters. An increase in residence time leads to an increase in recuperated heat. However, an increase in residence time also leads to an increase in the mass of catalyst in the reformer.

Table 4
Cost assumptions for economic evaluation of the TCR system.

Variable	Unit	Value
Natural gas	$/1000 m^3	250; 500
Catalyst	$/kg$_{cat}$	20; 50; 100
Electricity	$/MWh	150; 300
Catalyst replacement	month	3; 6; 9; 12; 15; 18; 21; 24

The economic effect from the use of a thermochemical recuperation system depends on the price of catalyst, electricity, and recuperated heat (in terms of natural gas cost). Table 4 summarizes the prices used in the economic analysis [46]. The economic evaluation was performed for a wide range of cost parameters because the price for natural gas and electricity varies throughout the year. Moreover, the price of catalyst depends on the type of catalyst. We analyzed the available prices of Ni-based catalyst [47]. In addition, the catalyst should be replaced from time to time to obtain a high level of methane conversion. The replacement period depends on many parameters. There is sufficient information in the literature that the replacement period can be from a few months to a few years [48]. We considered various replacement periods. In addition, we considered that the catalysts are fully replaced without regeneration.

Hourly net present profits for various cost parameters were calculated using Eq. (4) to determine the optimal parameters of the TCR system. Fig. 9 shows the dependence of net present profits due to using TCR on residence time and steam-to-methane ratio for various prices of catalyst and natural gas (upper charts – 250\$/1000 m^3; bottom

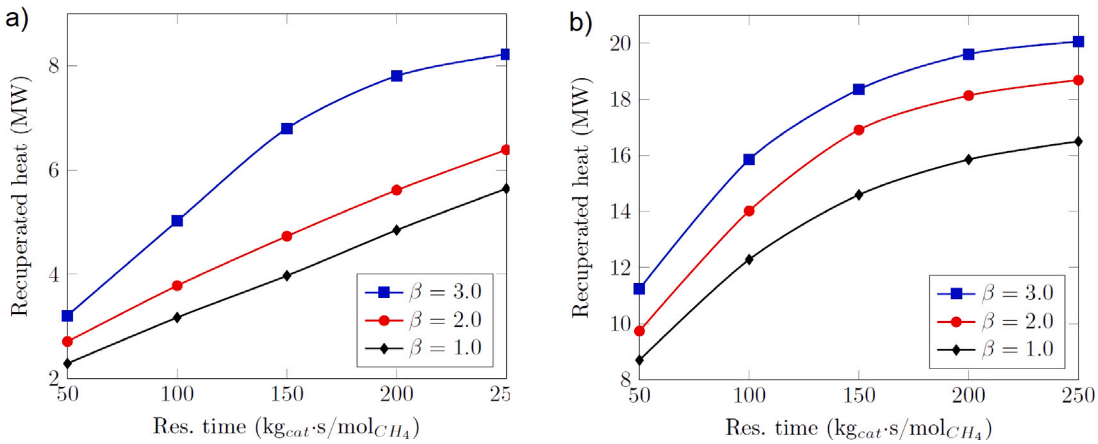

Fig. 8. CH_4 conversion as a function of residence time for various exhaust gas temperatures at reformer inlet: (a) T_{exh} = 900 °C; (b) T_{exh} = 1500 °C.

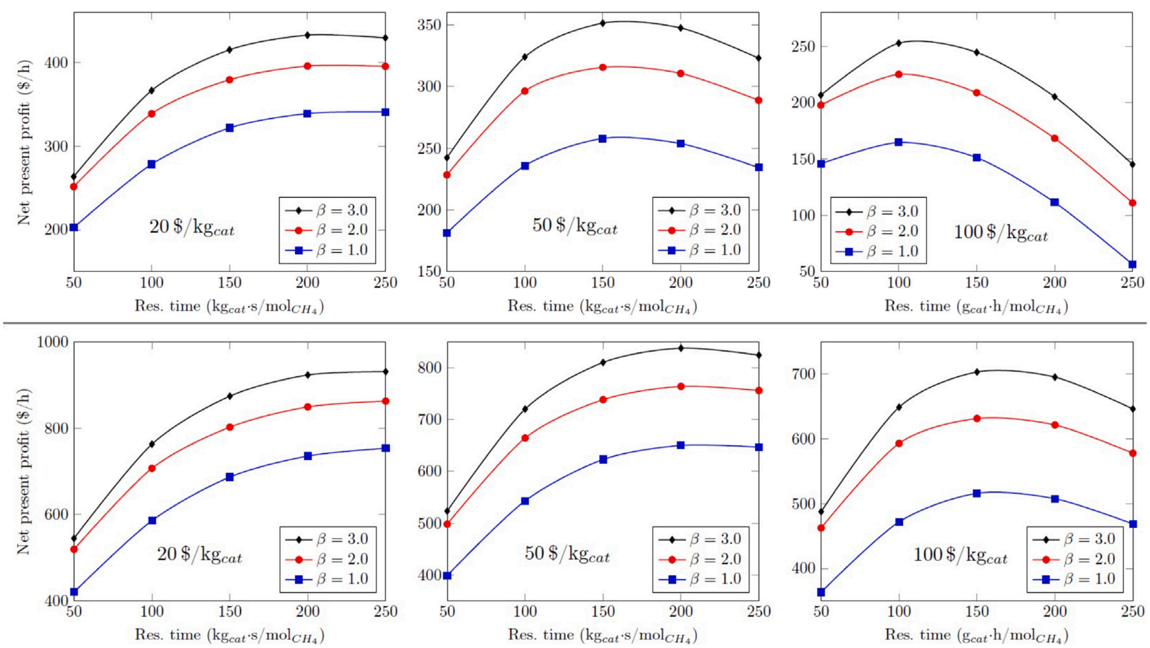

Fig. 9. Hour net present profit due to using thermochemical recuperation for T_{exh} = 1500 °C as a function of residence time and steam-to-methane ratio for various price of catalyst (20; 50; 100 $/kg): upper charts – price of natural gas is 250$/1000 m^3; bottom charts – price of natural gas is 500$/1000 m^3.

charts – 500$/1000 m^3). The calculations were performed for a catalyst replacement period of six months.

Fig. 9 shows that an increase in the price of catalysts leads to a decrease in the optimal residence time. For example, at a natural gas price of 250$/1000 m^3 the optimal residence time for 20$/$kg_{cat}$ is about 200–250 kg_{cat} s/mol_{CH_4}; for 50$/$kg_{cat}$ is about 150–200 kg_{cat} s/mol_{CH_4}; 100$/$kg_{cat}$ is about 100–150 kg_{cat} s/mol_{CH_4}. An increase in natural gas price leads to an increase in optimal residence time because a higher residence time provides higher recuperated heat and, as a result, higher net present profit. Moreover, Fig. 9 depicts that the economical efficiency of the use of thermochemical recuperation increases when the price of natural gas is growing.

The main goal of using the thermochemical waste-heat recuperation system is to save energy; therefore, optimizing operational parameters is a crucial task. For comparison, optimizing operational parameters for hydrogen production via steam methane reforming is not as crucial because the price of hydrogen is notably higher than the price of natural gas, and achieving maximal methane conversion is more important.

The economic efficiency of using TCR is also affected by the temperature of exhaust gas. Fig. 10 shows the effect of residence time and steam-to-methane ratio on our net present profit for various prices of catalyst and natural gas.

Fig. 10 shows that the absolute value of net present profit for T_{exh} = 900 °C is lower than that for T_{exh} = 1500 °C, for the same prices and operational parameters. The main reason for this observation is the lower recuperated heat for T_{exh} = 1500 °C, as the recuperated heat is mainly due to a change in enthalpy of synthesis gas, while the enthalpy of steam methane reforming is minimal. Fig. 10 illustrates that the optimal residence time can only be observed for a steam-to-methane ratio of 3.0, as the recuperated heat is maximal compared to steam-to-methane ratios of 1.0 and 2.0. Moreover, Fig. 10 shows that using TCR, even with heat recuperation, results in a negative net present profit because the cost of catalyst and electricity is higher than the saved money due to using TCR.

Comparing the data from Figs. 9 and 10, it can be concluded that TCR can provide maximum economic efficiency for high-temperature exhaust gas. This fact is confirmed by examples of industrial application of TCR, as most real TCR systems are designed for recuperation of high-temperature exhaust gas.

The net present profit also depends on the period of catalyst replacement. Fig. 11 shows the optimal residence time as a function of

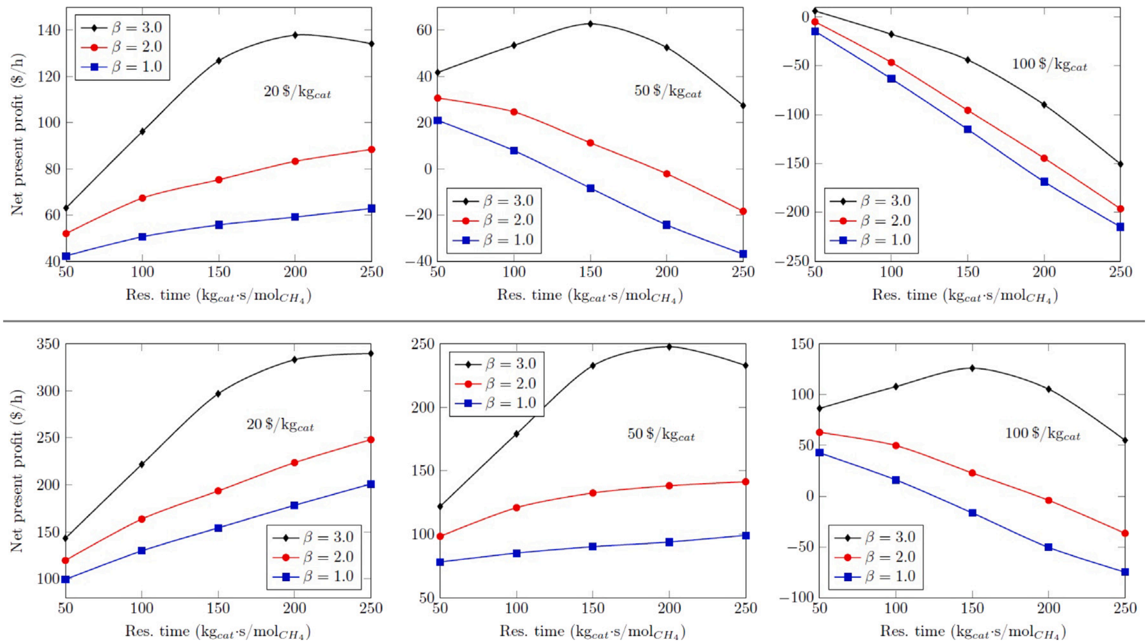

Fig. 10. Hour net present profit due to using thermochemical recuperation for T_{exh} = 900 °C as a function of residence time and steam-to-methane ratio for various price of catalyst (20; 50; 100 $/kg): upper charts – price of natural gas is 250$/1000 m^3; bottom charts – price of natural gas is 500$/1000 m^3.

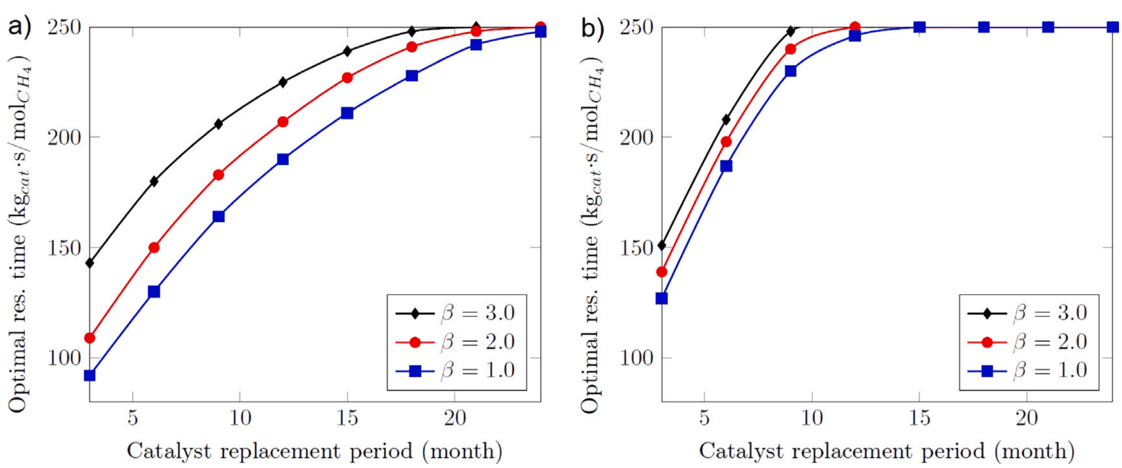

Fig. 11. Optimal residence time as function of catalyst replacement period for various steam-to-methane ratio and catalyst price for T_{exh} = 1500 °C: (a) natural gas price is 250$/1000 m^3; (b) natural gas price is 500$/1000 m^3.

catalyst replacement period for various operational and cost parameters for T_{exh} = 1500 °C.

Fig. 11 shows that an increase in the catalyst replacement period leads to an increase in optimal residence time. For a lower price of natural gas (250$/1000 m^3), the optimal residence time varies from 3 to 24 months because the short replacement period leads to an increase in costs for catalysts and, as a result, to a decrease in economic efficiency. The maximum residence time of 250 kg_{cat} s/mol_{CH_4} corresponds to close to equilibrium CH_4 conversion in a reformer. An increase in residence time above 250 kg_{cat} s/mol_{CH_4} will provide an increase in recuperated heat mainly due to a change in syngas enthalpy, and catalysts for higher residence time will have a lower effect on steam methane reforming enthalpy.

However, an increase in natural gas price leads to an increase in optimal residence time for a shorter period of catalyst replacement (Fig. 11). For a higher natural gas price (500$/1000 m^3), the maximal economic efficiency can be observed at a catalyst replacement period above 10–12 months. This fact confirms that the use of TCR is more efficient for the higher natural gas price.

The electricity consumption due to increased pressure drop in the thermochemical reformer was analyzed for two prices: 150 $/MWh and 300 $/MWh. The operational cost related to electricity is less than 2% in comparison to operational costs related to catalysts. Therefore, it can be concluded that increased electricity consumption has a negligible effect on operational cost balance.

The conducted calculations show that the economic efficiency of the use of thermochemical waste-heat recuperation systems depends on many parameters such as operational (exhaust gas temperature, steam-to-methane ratio), design (residence time, reformer tube dimensions), and cost (catalyst and natural gas price). Therefore, for the use of TCR in industrial gas furnaces, it is extremely important to consider all of those aspects because the main goal of the use of the TCR system is heat recuperation. For example, from a thermodynamic point of view, for T_{exh} = 1500 °C, an increase in residence time until the temperature of synthesis gas and exhaust gas at the reformer outlet will be equal, which will provide a positive energy effect [24]. However, the current techno-economic evaluation shows that an increase in residence time above the optimal one will have a negative effect on net present profit.

4. Conclusion

A techno-economic evaluation of a thermochemical waste-heat recuperation system based on steam methane reforming is presented. An operational cost analysis was conducted to understand the operational and price parameters that provide the maximum net present profit using the TCR system. The efficiency of the TCR system is characterized by recuperated heat, which was determined using a CFD model of an elementary reformer tube. The main results are the following:

- Recuperated heat for an exhaust gas temperature of 1500 °C is non-linearly increasing when the residence time (specific mass of catalyst per mole/s of inlet methane) increases. For example, for T_{exh} = 1500 °C and a steam-to-methane ratio of 3.0, the first 100 $kg_{cat}s/mol_{CH_4}$ of residence time provides about 16 MW of recuperated heat, while the last 100 $kg_{cat}s/mol_{CH_4}$ provides about 2 MW of recuperated heat.
- The net present profit had an extreme value that depended on operational and cost parameters for T_{exh} = 1500 °C. For T_{exh} = 900 °C and steam-to-methane ratios of 1.0 and 2.0, recuperated heat depends on residence time close to monotonic because in this case methane conversion is almost linearly dependent on residence time. As a result, no extremes were observed. However, for T_{exh} = 900 °C and a steam-to-methane ratio of 3.0, the dependence of the net present value on residence time shows extremes because in this case methane conversion at the reformer outlet is close to equilibrium at 250 $kg_{cat}s/mol_{CH_4}$ and recuperated heat as a function of residence time is close to logarithmic.
- The net present value also depends on the period of catalyst replacement. The optimal operation parameters to achieve the maximum net present value were determined for various catalyst replacement periods. Moreover, it was shown that additional electricity demand to cover the increased pressure drop in the recuperation system due to the use of the reformer has a negligible effect on operational costs.
- The results provide evidence that higher efficiency of the TCR system can be achieved for recuperation of high-temperature exhaust heat at high natural gas prices.

Nomenclature

C	Net profit value;
B	Cost of recuperated energy;
$OPEX_i$	ith operating expense;
H_{rec}	Recuperated heat;
LHV	Lower heating value;
H_i	Enthalpy;
ρ	Density;
v	Velocity;
T	Temperature;
p	Pressure;
$\overline{\overline{\tau}}$	Stress tensor;
$\rho\vec{g}$	The gravitational body force;
$\vec{F}$	External body forces;
S_i	Momentum source term for porous medium;
S_f^h	Energy source term (include the source of energy due to chemical reaction);
Y_j	Mass fraction of jth component of gas mixture;
$\vec{J}_j$	Diffusion flux of j component;
α	Permeability;
C_2	Inertial resistance factor;
μ	Viscosity;
D_p	Catalyst particle diameter;
ϵ_{ip}	Packed bed porosity;
$k_{\mathrm{eff,f,s}}$	Effective thermal conductivity in the porous medium, fluid, solid phases, respectively;
R_i	Reaction rate of ith reaction;
η	Effectiveness factor;
M_{cat}	Mass of catalyst;
γ	Correlation factor;
TCR	Thermochemical recuperation;
CFD	Computational fluid dynamics;
SMR	Steam methane reforming;
exh	Exhaust heat (gas);
cat	Catalyst.

CRediT authorship contribution statement

Dmitry Pashchenko: Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Software, Resources, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Igor Karpilov:** Writing – review & editing, Software, Investigation, Formal analysis. **Mikhail Polyakov:** Writing – review & editing, Software, Methodology, Investigation. **Stanislav K. Popov:** Writing – review & editing, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No data was used for the research described in the article.

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